SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Sameer Ahmed G	**Reg. No.:**	**192421154**
Section / Batch:	**B Tech IT**	**Date:**	**27-07-2026**

Instructions to Students

- Answer the following question. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1

SECTION A – CONCEPT MAPPING

Code of Conduct

I Sameer Ahmed G (192421154) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sameer Ahmed G

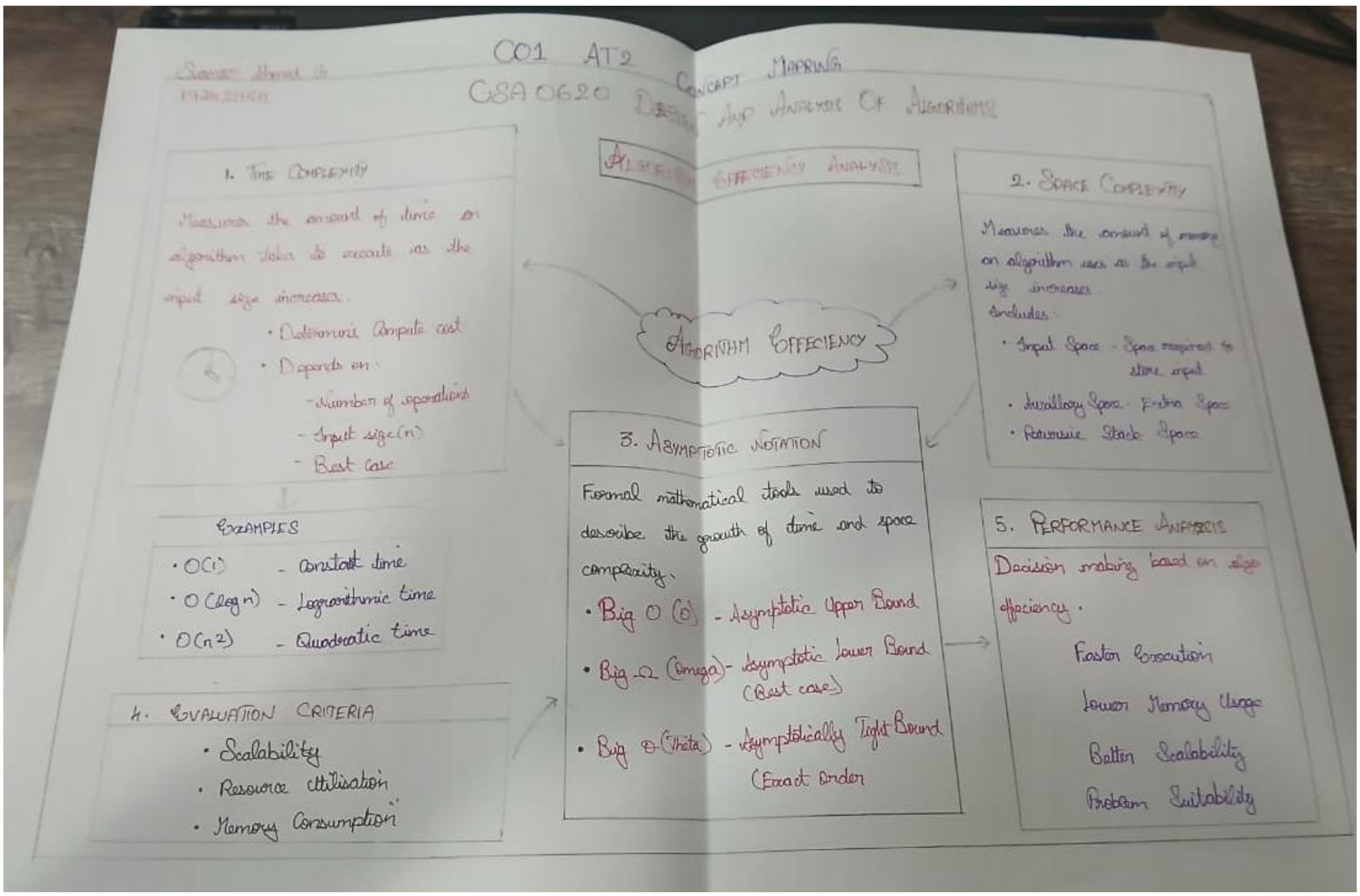
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%				
Linkages	25%				
Organization	20%				
Integration of Literature	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________



Complete View of Algorithm Performance

Time Complexity + Space Complexity = Complete Evaluation

Common Time Compexity (Increasing Order)

$O(1)$ Constant	$\Theta(\log n)$ Logarithmic	$O(n)$ Linear	$O(n \log n)$ Linearithmic

Best → Worse

$\Theta(n!)$ Factorial	$\Theta(2^{n2})$ Exponential	$\Theta(n^2)$ Quadratic

Worse ← Best

Algorithm Performance

Types of Space

Input Space – Memory needed to store the input data

Auxiliary Space – Extra memory used by the algo during execution

Stack Space – Memory Used by Recursive Calls.

Trade Off Between Time And Space

Focus On Time	Focus On Space
Faster Execution but may require more memory	Less memory usage but may take more time

Choosing Right balance is the key here

Conclusion

Balanced Time and Space Comp leads to efficient, Scalable and Practical Algorithms.

Real World Applications

- Sorting Algorithms
- Searching Algorithms
- Graph Algorithms
- Dynamic Programming
- Operating Systems